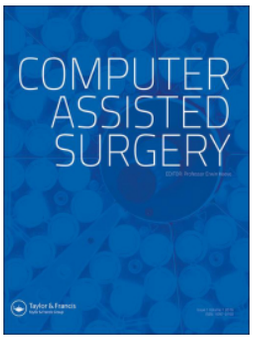


Computer Assisted Surgery

ISSN: 2469-9322 (Online) Journal homepage: www.tandfonline.com/journals/icsu21

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**To cite this article:** Xin Wang , Wenmao Yan , Jingdong Shi, Shi Cheng, Wei Yu & Hongyi Zhang (2025) Exploring the survival benefits of surgical treatment for pancreatic adenocarcinoma using the DeepSurv neural network model, Computer Assisted Surgery, 30:1, 2556334, DOI: 10.1080/24699322.2025.2556334

**To link to this article:** https://doi.org/10.1080/24699322.2025.2556334

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COMPUTER ASSISTED SURGERY
2025, VOL. 30, NO. 1, 2556334
https://doi.org/10.1080/24699322.2025.2556334

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# Exploring the survival benefits of surgical treatment for pancreatic adenocarcinoma using the DeepSurv neural network model

Xin Wang, Wenmao Yan, Jingdong Shi, Shi Cheng, Wei Yu and Hongyi Zhang

Department of General Surgery, Beijing Tiantan Hospital, Capital Medical University, Beijing, China

**ABSTRACT**

To develop a DeepSurv model for predicting survival in pancreatic adenocarcinoma patients, evaluating the benefit of surgical versus non-surgical treatment across different stages, including stage IV subcategories. Clinical data were extracted from the SEER database (2000–2020). Patients were randomly divided into a model-building group and an experimental group. The DeepSurv model was trained and hyperparameter-optimized. Simulated paired data were created by switching treatment status. Predicted survival rates were compared using generalized estimating equations. SHAP values analyzed variable importance.The study included 16,068 patients. The final model achieved a C-index of 0.85. Surgical treatment yielded higher survival rates than non-surgical across all stages ($p<0.001$), though the benefit diminished in advanced stages. For stage IV, surgery improved survival in T1–3 and N0 stages ($p<0.001$) but not in T4 and N1. SHAP analysis ranked M stage as the most significant predictor of mortality, followed by T stage, overall stage, and surgical status. M1 metastasis was associated with a 14% increased mortality risk, while surgery reduced risk by 11%. Surgery reduces mortality across stages, with declining efficacy in advanced disease. For stage IV patients, surgery is beneficial except for those with T4 or N1 disease. Combining DeepSurv with SHAP analysis facilitates individualized prediction of surgical survival benefits.

**KEYWORDS**
Pancreatic adenocarcinoma; survival prediction; DeepSurv; primary tumor resection

## 1. Introduction

Pancreatic cancer (PC) is a highly aggressive malignant tumor with a very poor prognosis [1]. It is currently the seventh leading cause of cancer death worldwide, with a 5-year survival rate as low as 3–15% [2–4]. Surgery is the only curative treatment for pancreatic cancer [5] and can potentially increase the 5-year survival rate to 20% [6,7]. However, the resection rate is very low, only 20–25% [8], because ~60% of newly diagnosed PC cases are metastatic [5,9–11]. For these patients, the National Comprehensive Cancer Network (NCCN) and the American Society of Clinical Oncology (ASCO) do not recommend resection of the primary tumor [12,13]. However, there is ongoing debate about whether primary tumor resection (PTR) or combined synchronous metastasectomy should be performed in stage IV PC [14–22]. Some studies suggest that surgical resection is beneficial when patients are appropriately selected [15,18,23,24]. In contrast, other literature indicates that resection of the tumor and its metastatic sites does not confer a survival benefit and is therefore not recommended [25–27].

Artificial intelligence (AI), particularly neural networks within deep learning, has been widely applied in the medical field [28]. Neural networks are a branch of machine learning that utilize nonlinear models composed of a set of highly interconnected processing units (neurons) and weighted connections [29]. Currently, neural network models are widely used for predicting medical outcomes. Approximately 78% of studies indicate that neural network models improve treatment decision-making

**CONTACT** Hongyi Zhang zhyl3910433778@163.com Department of General Surgery, Beijing Tiantan Hospital, Capital Medical University, Beijing, China

Supplemental data for this article is available online at https://doi.org/10.1080/24699322.2025.2556334

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compared to existing statistical methods, while the remaining 22% perform at least as well as existing models [30].

The DeepSurv model, developed by Katzman et al. [31] is a neural network model for survival analysis that applies deep learning techniques to nonlinear Cox regression. It uses clinical data as input and optimizes the model through hyperparameters like dropout, learning rate decay, and regularization to predict survival rates at different times.

Establishing a highly performant survival prediction model aids doctors in recommending the most appropriate treatment plans to maximize survival benefits. Therefore, this study constructs a DeepSurv neural network survival analysis model to predict the survival rates of PC patients at different stages and within various T and N stages in stage IV under surgical and non-surgical treatments, thereby evaluating the survival benefits of surgical treatment.

## 2. Materials and methods

### 2.1. Data source

Using SEER*stat software (version 8.3.5), cases from January 2000 to December 2020 covering 17 regions in the United States were selected.

### 2.2. Inclusion/exclusion criteria

Inclusion criteria were based on the 3rd edition of the International Classification of Diseases for Oncology (ICD-O-3) for the pancreas (C25.0-C25.9), with histological codes for adenocarcinoma (8140/3), age ≥18 years. Collected variables included gender, age, race, TNM stage (AJCC 7th edition), tumor size, tumor location, surgical status, survival time, and cause of death. Exclusion criteria included incomplete clinical information, missing survival time and follow-up data, incomplete demographic information, and non-pancreatic cancer-related deaths. The primary endpoint was the time from initial diagnosis to pancreatic cancer-related death or end of follow-up.

### 2.3. Data grouping

Cases with different TNM stages were randomly divided using the train_test_split module from Python's scikit-learn package in a 9:1 ratio, forming the model group and TNM experimental group. The model group was used for model construction, while the TNM experimental group was used for prediction analysis. The model group was further randomized into training and validation groups in an 8:2 ratio. The TNM experimental group was divided into surgical and non-surgical groups. Simulated paired data were created by converted surgical cases to non-surgical and vice versa, forming the TNM paired group. The same method was used to construct the stage IV T and N experimental and paired groups.

### 2.4. Data preprocessing

For the model cohort data, the SMOTE module from the imblearn package in Python 3.7 was employed to address class imbalance based on mortality outcomes. Specifically, this was implemented by applying SMOTE exclusively to the baseline covariates of uncensored deceased samples, thereby preserving the integrity of survival time labels. The ColumnTransformer StandardScaler module from scikit-learn was used to normalize the data, ensuring equal treatment of each feature by the classifier.

### 2.5. Variable selection

Feature selection was based on univariate and multivariate Cox regression analyses, combined with clinical experience to select appropriate variables for model construction.

### 2.6. The DeepSurv model architecture

The DeepSurv algorithm constitutes a Cox proportional hazards model based on feed-forward neural networks, comprising an input layer, multiple hidden layers, and an output layer. Its loss function integrates the Cox proportional hazards loss (negative log partial likelihood) augmented with regularization terms, enabling survival outcome prediction and individualized survival probability estimation. We separately constructed models using both raw and balanced data. Bayesian hyperparameter optimization [32] was implemented to select critical parameters including learning rate, learning rate decay, L1/L2 regularization coefficients, dropout rates, and activation functions. Specific architectural details of the DeepSurv model are provided in the Supplementary Material (see Supplementary Material). Model performance was ultimately evaluated using the concordance index on validation cohorts, supplemented by Brier score for calibration assessment and SHAP values for variable importance analysis.

### 2.7. Statistical analysis

Continuous data were expressed as mean ± standard deviation ($x \pm s$); categorical data were expressed as $n$ (%), with $p < 0.05$ indicating statistical significance. The TFDeepSurv package in Python 3.7 was used to construct the DeepSurv model. Model performance was evaluated using the C-index and Brier score. The matching quality between the experimental and matched groups was assessed using standardized mean differences (SMD). An SMD threshold exceeding 0.1 indicated insufficient balance, necessitating refinement through propensity score matching to harmonize baseline characteristics. Due to non-normal distributions and strong interdependencies between the two groups, survival rate outcomes were evaluated *via* a generalized estimating equations (GEE). GEE implemented in Python 3.7 with the statsmodels library (version 0.14.4). An exchangeable working correlation structure was specified to account for intra-group dependencies, while the robust sandwich variance estimator was employed to ensure the validity of parameter estimates under potential misspecification of the correlation structure. Statistical significance was determined at $p < 0.05$, with an adjusted $p$-value threshold of 0.025 ($\alpha = 0.025$) accounting for multiple comparisons.

## 3. Results

### 3.1. Data information

A total of 127,605 pancreatic cancer cases were retrieved from the SEER database, with 16,068 cases meeting the inclusion criteria. Figure 1 shows the screening procedure applied to identify patients with pancreatic adenocarcinoma.

The average age was 68.3 ± 12.3 years, with 40,596 males (50.8%) and 64,377 white patients (80.6%). The number of deaths was 72,876 (91.2%), with an average survival time of 15.0 ± 24.2 months. There were 19,200 surgical cases (24.0%), and the average tumor size was 42.0 ± 30.9 mm. The basic clinical characteristics of the alive and deceased groups are listed in Table 1, with significant differences ($p < 0.05$) in all variables except N stage.

### 3.2. Variable selection results

The results of univariate and multivariate analyses are presented in Table 2. The findings indicate that race, tumor location, and T and N staging do not have a statistically significant impact on survival time. Based on clinical experience, all variables except for race were included in the model.

### 3.3. The DeepSurv model results

A DeepSurv model was established for the data, configured with an input layer of 8 nodes, two hidden layers (the first hidden layer with 7 nodes, and the second hidden layer with 3 nodes), and an output layer with 1 node. Based on the results of Bayesian hyperparameter optimization, the learning rate was set to 0.01, the learning rate decay to 0.01, the activation function to tanh, the dropout to 0.3, the

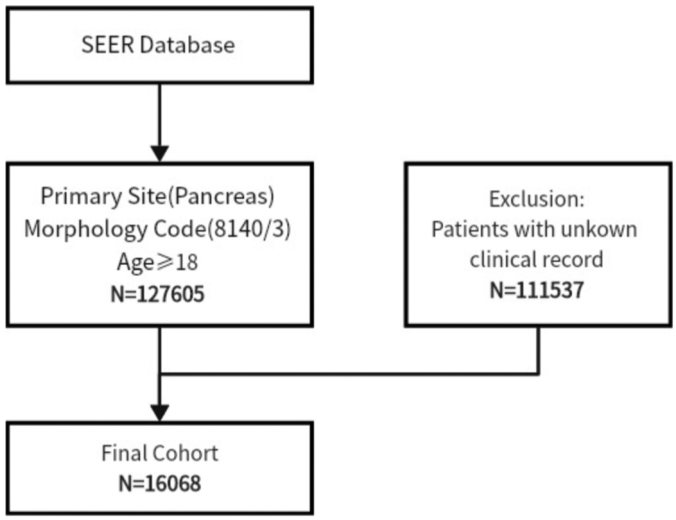


**Figure 1.** The flow diagram of patients with pancreatic adenocarcinoma.

**Table 1.** Baseline data of patients.

| | *e* | | |
|---|---|---|---|
| Characteristic | Alive, $N=765^a$ | Death, $N=15{,}303^a$ | *p*-Value[b] |
| Size | 33 ± 19 | 42 ± 32 | <0.001 |
| Survival time (months) | 75 ± 36 | 11 ± 13 | <0.001 |
| Sex | | | 0.046 |
| Male | 374 (49%) | 8047 (53%) | |
| Female | 391 (51%) | 7256 (47%) | |
| Race | | | 0.002 |
| White | 584 (76%) | 12,369 (81%) | |
| Others | 181 (24%) | 2934 (19%) | |
| Age | | | <0.001 |
| <60 | 256 (33%) | 3500 (23%) | |
| ≥60 | 509 (67%) | 11,803 (77%) | |
| Location | | | <0.001 |
| Head | 503 (66%) | 8116 (53%) | |
| BodyTail | 173 (23%) | 4711 (31%) | |
| Others | 89 (12%) | 2476 (16%) | |
| Stage | | | <0.001 |
| IA | 68 (9%) | 159 (1%) | |
| IB | 72 (9%) | 748 (5%) | |
| IIA | 203 (27%) | 1860 (12%) | |
| IIB | 254 (33%) | 2710 (18%) | |
| III | 57 (7%) | 1591 (10%) | |
| IV | 111 (15%) | 8235 (54%) | |
| T stage | | | <0.001 |
| T1 | 83 (11%) | 561 (4%) | |
| T2 | 136 (18%) | 4377 (29%) | |
| T3 | 462 (60%) | 6955 (45%) | |
| T4 | 84 (11%) | 3410 (22%) | |
| N stage | | | 0.64 |
| N0 | 443 (58%) | 8732 (57%) | |
| N1 | 322 (42%) | 6571 (43%) | |
| M stage | | | <0.001 |
| M0 | 654 (85%) | 7068 (46%) | |
| M1 | 111 (15%) | 8235 (54%) | |
| Surgery | | | <0.001 |
| Surgery | 568 (74%) | 3225 (21%) | |
| Unsurgery | 197 (26%) | 12,078 (79%) | |

T stage: tumor stage; N stage: regional lymph node stage; M stage: metastasis stage.
[a]Mean ± *SD*; *n* (%).
[b]Welch two sample *t*-test; Pearson's chi-squared test.

optimizer to adam, the L1 regularization value to 2.13e-5, and the L2 regularization value to 8.07e-4. The final model demonstrated discriminative performance with concordance indices (C-index) of 0.85 in both training and testing sets. Calibration precision was evidenced by a Brier score of 1.01 e-3. SHAP value analysis (Figure 2) and SHAP global feature importance ranking (Table 3) revealed metastatic stage (M stage) as the most salient prognostic determinant, demonstrating an inverse relationship with survival rates. Subsequent influential predictors included tumor invasion depth (T stage), overall tumor stage, and surgical intervention status, whereas racial background exhibited minimal prognostic impact. Surgical status ranked prominently in global SHAP importance, though individual-level predictive contributions may exhibit case-specific heterogeneity (see Supplementary Figures 1–3). Model analysis revealed distant metastasis (M1) increased mortality risk by 14% (HR = 1.14, 95%CI: 1.02–1.27), while surgical intervention reduced risk by 11% (HR = 0.89, 95%CI: 0.82–0.96). Although statistically significant, clinical relevance must be interpreted in conjunction with other prognostic factors. Surgery appeared among the top three predictive features with a frequency of 0.3224 (32.24%), though cancer stage and metastasis status (M) demonstrated greater importance for high-risk prediction. These results reflect the final DeepSurv model trained on normalized, SMOTE-balanced data with optimized hyperparameters.

### 3.4. Predicted survival rate results

The standardized mean difference (SMD) between the TNM experimental cohort and the paired group was calculated as 0.0037, indicating sufficient matching balance (SMD < 0.1 threshold). Formal *E*-value analysis estimates that unobserved confounding factors would need to exhibit substantial association strengths (relative risk >2.8) to nullify the identified surgical benefits. The sensitivity analysis for propensity score matching yielded similar results, with $\Gamma$ critical = 2.76. The TNM experimental group and the

Table 2. Results of univariate and multivariate analyses.

| | Univariable | | | | | Multivariable | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Characteristic | *N* | Event *N* | HR | 95% CI | *p*-Value | *N* | Event *N* | HR | 95% CI | *p*-Value |
| Sex | | | | | | | | | | |
| Male | 8421 | 8047 | — | — | | 8421 | 8047 | — | — | |
| Female | 7647 | 7256 | 0.93 | 0.90, 0.96 | <0.001 | 7647 | 7256 | 0.94 | 0.91, 0.97 | <0.001 |
| Race | | | | | | | | | | |
| White | 12,953 | 12,369 | — | — | | 12,953 | 12,369 | — | — | |
| Others | 3115 | 2934 | 1.04 | 1.00, 1.08 | 0.082 | 3115 | 2934 | 1.06 | 1.01, 1.10 | 0.009 |
| Age | | | | | | | | | | |
| <60 | 3756 | 3500 | — | — | | 3756 | 3500 | — | — | |
| ≥60 | 12,312 | 11,803 | 1.27 | 1.23, 1.32 | <0.001 | 12,312 | 11,803 | 1.35 | 1.30, 1.40 | <0.001 |
| Location | | | | | | | | | | |
| Head | 8619 | 8116 | — | — | | 8619 | 8116 | — | — | |
| BodyTail | 4884 | 4711 | 1.41 | 1.36, 1.46 | <0.001 | 4884 | 4711 | 1.02 | 0.98, 1.06 | 0.371 |
| Others | 2565 | 2476 | 1.38 | 1.32, 1.44 | <0.001 | 2565 | 2476 | 1.05 | 1.00, 1.10 | 0.037 |
| Stage | | | | | | | | | | |
| IA | 227 | 159 | — | — | | 227 | 159 | — | — | |
| IB | 820 | 748 | 2.39 | 2.01, 2.83 | <0.001 | 820 | 748 | 1.57 | 1.29, 1.92 | <0.001 |
| IIA | 2964 | 2710 | 1.79 | 1.53, 2.10 | <0.001 | 2964 | 2710 | 1.94 | 1.60, 2.35 | <0.001 |
| IIB | 2063 | 1860 | 1.95 | 1.66, 2.30 | <0.001 | 2063 | 1860 | 1.74 | 1.43, 2.11 | <0.001 |
| III | 1648 | 1591 | 2.67 | 2.27, 3.15 | <0.001 | 1648 | 1591 | 1.68 | 1.38, 2.05 | <0.001 |
| IV | 8346 | 8235 | 5.06 | 4.32, 5.92 | <0.001 | 8346 | 8235 | 2.74 | 2.27, 3.30 | <0.001 |
| T stage | | | | | | | | | | |
| T1 | 644 | 561 | — | — | | 644 | 561 | — | — | |
| T2 | 4513 | 4377 | 1.91 | 1.75, 2.09 | <0.001 | 4513 | 4377 | 1.09 | 0.98, 1.21 | 0.100 |
| T3 | 7417 | 6955 | 1.13 | 1.04, 1.24 | 0.004 | 7417 | 6955 | 0.96 | 0.86, 1.06 | 0.411 |
| T4 | 3494 | 3410 | 1.60 | 1.47, 1.75 | <0.001 | 3494 | 3410 | 0.93 | 0.84, 1.04 | 0.218 |
| N stage | | | | | | | | | | |
| N0 | 9175 | 8732 | — | — | | 9175 | 8732 | — | — | |
| N1 | 6893 | 6571 | 0.87 | 0.85, 0.90 | <0.001 | 6893 | 6571 | 1.07 | 1.03, 1.11 | 0.002 |
| M stage | | | | | | | | | | |
| M0 | 7722 | 7068 | — | — | | 7722 | 7068 | — | — | |
| M1 | 8346 | 8235 | 2.51 | 2.43, 2.59 | <0.001 | 8346 | 8235 | | | |
| Surgery | | | | | | | | | | |
| Surgery | 12,275 | 12,078 | — | — | | 12,275 | 12,078 | — | — | |
| Unsurgery | 3793 | 3225 | 0.30 | 0.29, 0.31 | <0.001 | 3793 | 3225 | 0.37 | 0.35, 0.39 | <0.001 |
| Size | 16,068 | 15,303 | 1.00 | 1.00, 1.00 | <0.001 | 16,068 | 15,303 | 1.00 | 1.00, 1.00 | <0.001 |

HR: hazard ratio; CI: confidence interval; T stage: tumor stage; N stage: regional lymph node stage; M stage: metastasis stage.

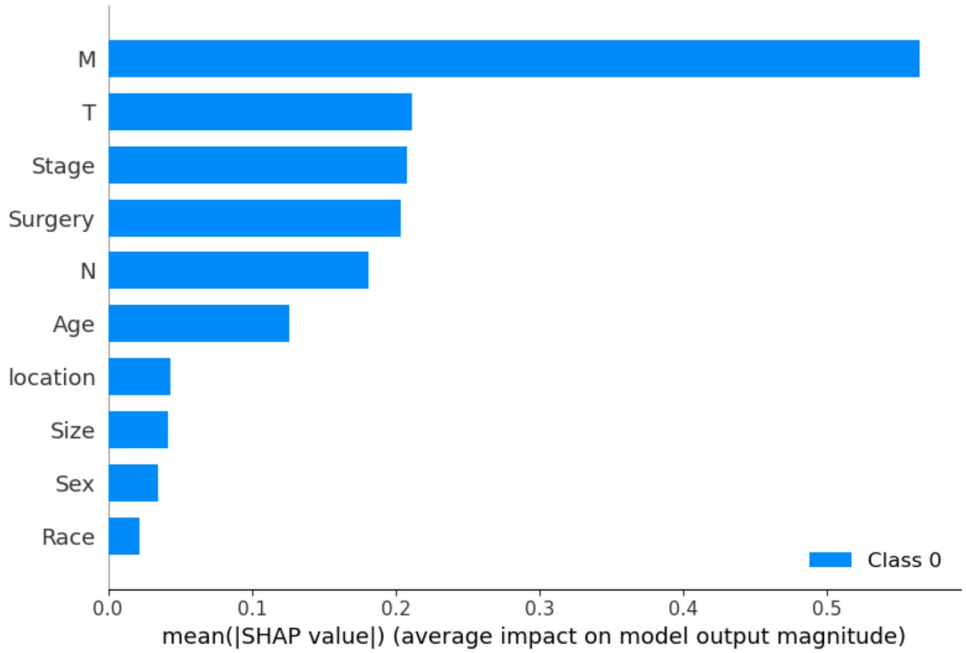


Figure 2. SHAP value analysis.

Table 3. SHAP global feature importance ranking.

| Feature | Importance |
|---|---|
| M | 0.571959 |
| Stage | 0.233281 |
| Surgery | 0.211851 |
| N | 0.207636 |
| T | 0.183272 |
| Age | 0.155370 |
| Location | 0.053921 |
| Size | 0.049164 |
| Sex | 0.041420 |
| Race | 0.035347 |

paired group were input into the model to obtain predicted survival rates. The predicted average 5-year survival rates for surgical and non-surgical treatments across different stages are shown in Figure 3.

Generalized additive model (GAM) analysis revealed that stage-stratified experimental cohorts demonstrated significantly higher survival rates compared to their matched counterparts ($p < 0.001$). The T stage experimental group and paired group of stage IV were input into the model to obtain the corresponding 5-year survival rates. The average survival rates are shown in Figure 4.

Survival rate predictions derived from generalized estimating equations(GEE)analyses yielded statistically significant associations ($p < 0.001$) across all subgroups, with the exception of the stage IV T4 cohort (the Bonferroni-corrected $p = 0.12$). The N stage experimental group and paired group of stage IV were input into the model to obtain the corresponding 5-year survival rates. The average survival rates are shown in Figure 5.

Survival rate predictions were evaluated through generalized estimating equations(GEE) analyses, revealing statistically significant associations ($p < 0.001$) in the stage IV N0 cohort, while no statistically significant association was observed in the N1-positive subgroup (the Bonferroni-corrected $p = 0.44$).

Multivariable Cox regression confirmed survival benefits from surgery across all subgroups. In the overall cohort, surgery significantly reduced mortality risk (HR = 0.44, 95%CI: 0.42–0.45, $p < 0.0001$). Among Stage IV patients, surgery decreased mortality risk by 50% (HR = 0.50, 95%CI: 0.44–0.57, $p < 0.0001$). Stratified analysis by T/N stage revealed:

Stage IV T4 patients: HR = 0.48 (95%CI: 0.42–0.55, $p < 0.0001$)

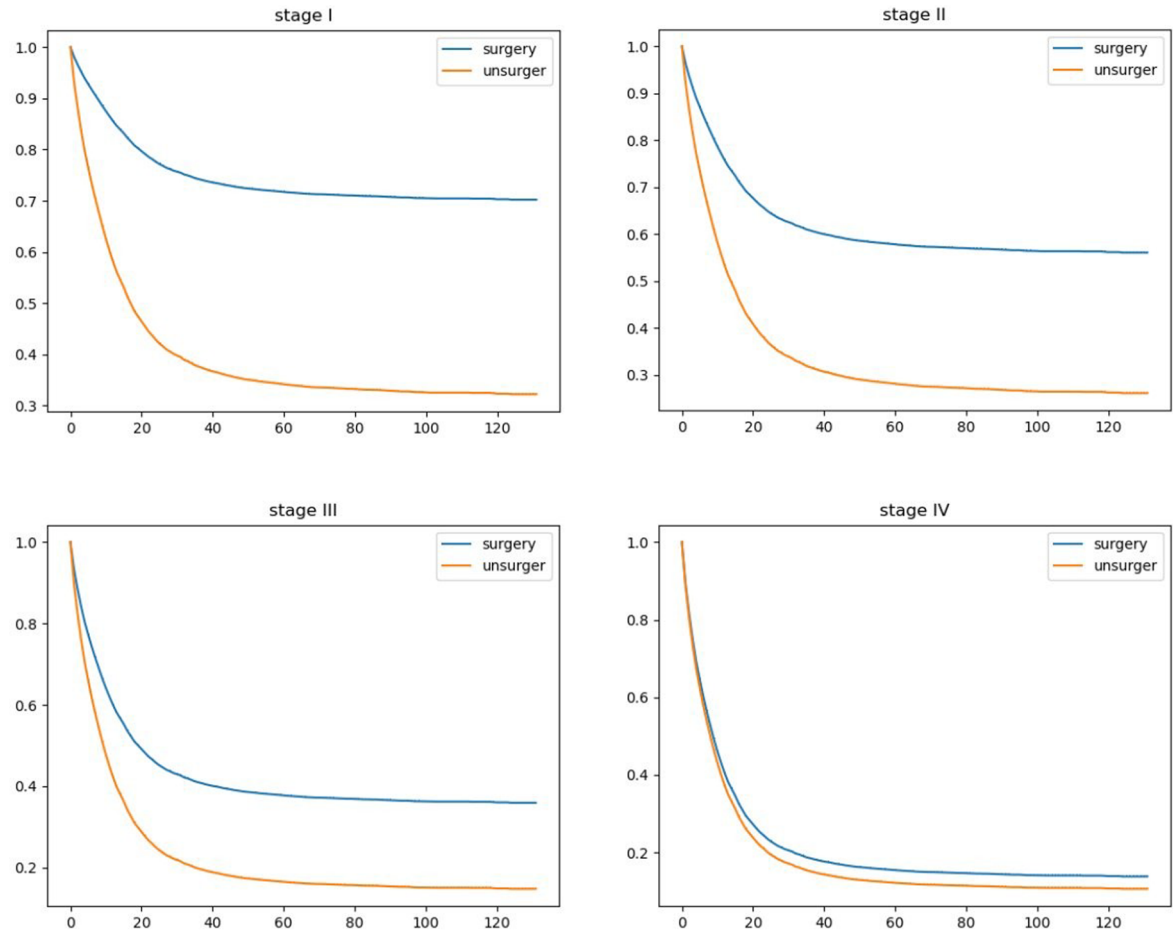


Figure 3. Average 5-year survival rates for surgical and non-surgical treatments across different stages.

Stage IV N0 patients: HR = 0.37 (95%CI: 0.30–0.45, $p < 0.0001$)
Stage IV N1 patients: HR = 0.66 (95%CI: 0.55–0.79, $p < 0.0001$).

The original Cox model demonstrated excellent discriminative ability (Concordance = 0.83) and confirmed the protective effect of surgery (HR = 0.44, 95%CI: 0.42–0.45).

## 4. Discussion

The surgical treatment of pancreatic cancer is considered the primary therapeutic approach recommended by the NCCN and the American Joint Committee on Cancer (AJCC) for patients with stage I and II (localized and resectable) pancreatic cancer [5,13], offering favorable prognostic outcomes [33]. Studies indicate that patients with stage I pancreatic cancer who do not undergo pancreatic resection have a median survival period of 8 months [34], whereas those undergoing surgical intervention achieve longer survival durations [35].

According to the NCCN guidelines, unlike hepatectomy for liver metastases commonly used in colorectal cancer, tumor resection should not be performed when tumors have distant metastases [5,13,36]. However, current research indicates an increasing number of stage IV pancreatic cancer patients have resectable primary and metastatic lesions [37].

However, the question of whether surgical treatment should be offered to stage IV patients remains controversial, with most studies suggesting that surgery can still confer survival benefits for stage IV patients [14,15,18,23,26,38]. Research indicates that surgery can prolong Overall Survival (OS) in patients with metastatic pancreatic cancer (mPC) [14,15,23,39], particularly in those with oligometastatic disease, with increases in survival time reaching up to 5.5 months [38]. In a study by Kim et al. [39], 115 patients diagnosed with mPC during surgery included 35 who underwent primary tumor resection (PTR). Results

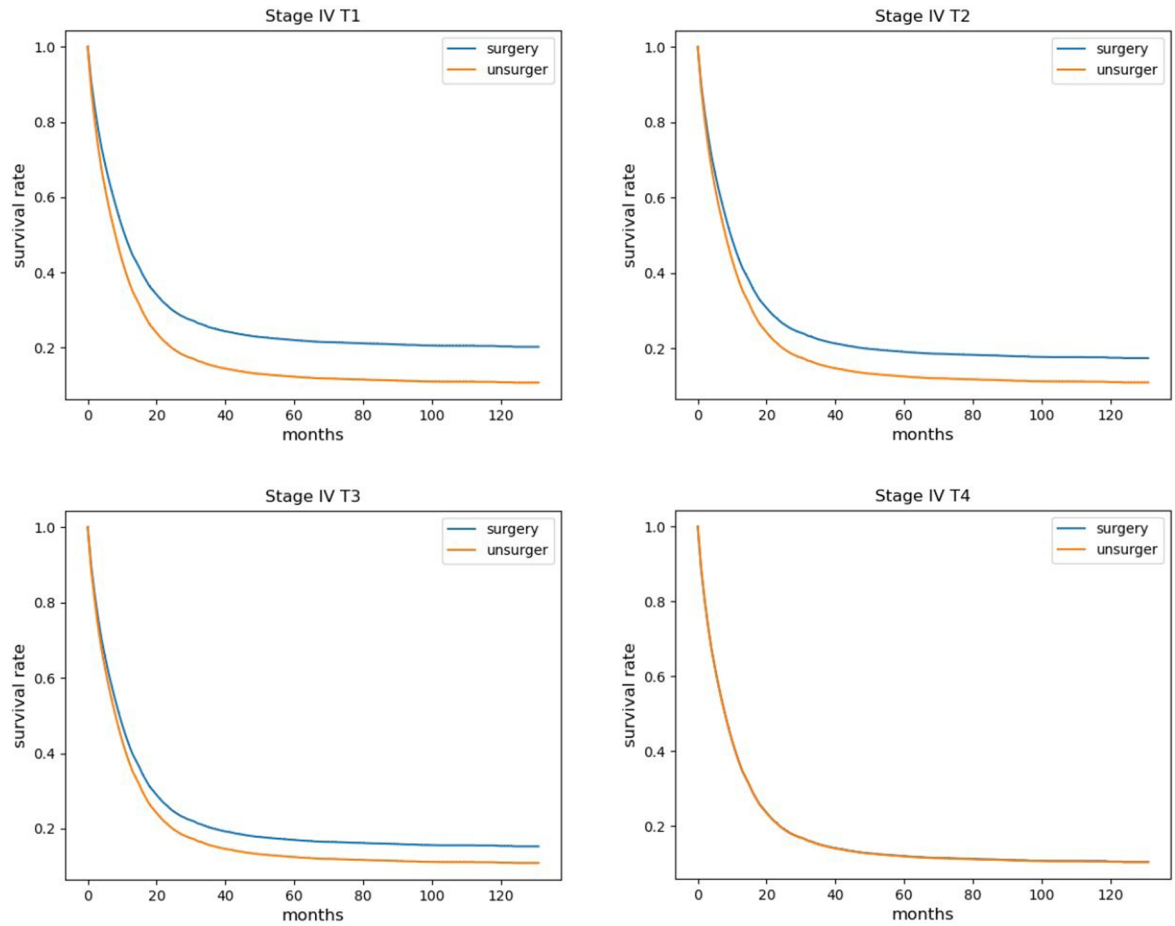


Figure 4. Average 5-year survival rates for surgical and non-surgical treatments across different T stages in stage IV.

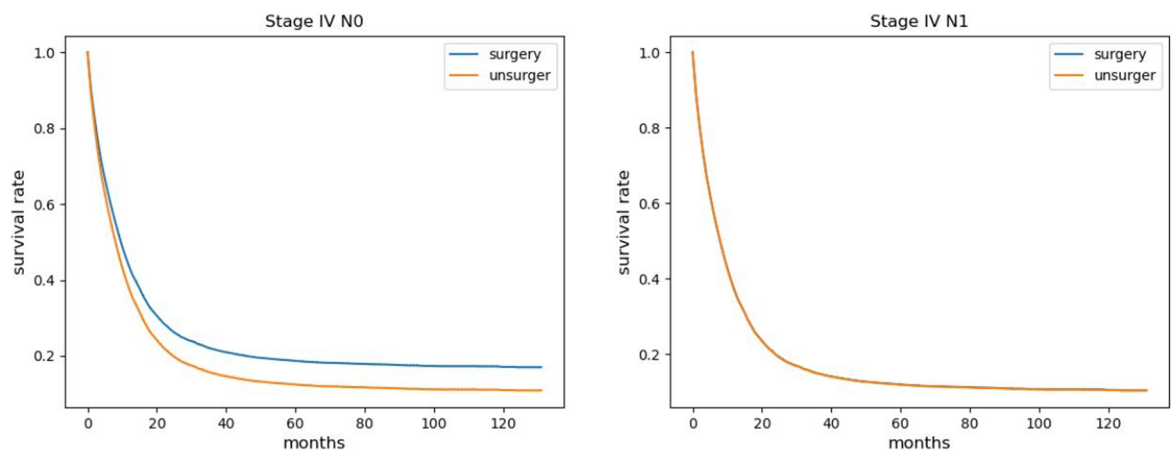


Figure 5. Average 5-year survival rates of surgical and non-surgical treatments for different N stages of stage IV pancreatic cancer.

demonstrated significantly improved survival rates among resected patients compared to those who did not undergo resection.

Current literature suggests that simultaneous or staged resection of metastatic lesions may be appropriate for patients with metastatic pancreatic cancer (mPC) [18,20,23,39–43]. According to previous studies, perioperative mortality rates for pancreatic cancer (PC) resection combined with synchronous metastasis resection can be as low as 0% [40,42], indicating that surgery is safe and feasible, potentially providing

survival benefits for specific patients [41]. Tachezy et al. [18] reported that patients undergoing simultaneous resection of the pancreatic primary and liver metastases had longer overall survival compared to non-resected patients (median overall survival 14 *vs.* 8 months, $p < 0.001$). In a study by Hamad et al. [20], which included 47,785 mPC patients with isolated liver metastases, patients were divided into non-surgical ($n = 46{,}894$) and surgical ($n = 891$) groups. Results showed that the median overall survival in the surgical group was significantly higher than in the non-surgical group (10.74 *vs.* 3.4 months, $p < 0.001$). However, some scholars strongly oppose extensive resection in cases of liver metastases. Their research indicates that not only does overall survival not improve, but surgery-related morbidity and mortality rates also increase [27].

From the perspective of tumor molecular biology, pancreatic cancer is characterized as a tumor with a dense stroma and a tough texture, making it difficult for chemotherapy drugs to penetrate into the tumor, resulting in poor treatment responses. Moreover, the stroma also promotes tumor survival, proliferation, and metastasis [44,45]. Therefore, surgical resection of the tumor helps reduce tumor burden and extends overall survival [14].

Based on the aforementioned debates, this study employs artificial intelligence techniques to construct a neural network survival analysis model for predicting survival rates. Numerous studies have shown that neural networks outperform traditional logistic regression algorithms in predicting survival rates accurately [46–49]. The model can also provide individual survival probabilities at various time points, an aspect that, to our knowledge, has not been explored in similar studies. In clinical practice, this model can be used for personalized patient prognosis.

Compared to conventional predictive models, such as Cox proportional hazards models and parametric survival models, which require strict adherence to underlying assumptions (e.g. proportional hazards, linearity) and may introduce bias when these assumptions are violated, neural network models circumvent these limitations by automatically learning nonlinear feature interactions through multilayer architectures without requiring predefined functional forms. Traditional models typically output hazard ratios (HRs) or population-averaged survival curves, facilitating clinical interpretation of variable effects, but they struggle to capture complex nonlinear relationships unless manually specified *via* polynomial transformations. In contrast, neural networks enhance predictive accuracy through nonlinear modeling and enable individualized risk stratification by estimating personalized risk scores under different interventions, thereby supporting optimized therapeutic recommendations. However, the intrinsic opacity of neural network decision logic—often termed the 'black box' dilemma—may undermine clinical trustworthiness, necessitating integration of explainability tools like SHAP and LIME frameworks. Additionally, neural networks demand substantial computational resources and large sample sizes (typically $*n* > 5000$) for robust training, which may constrain clinical applicability compared to traditional methods.

The DeepSurv algorithm represents a survival analysis framework grounded in deep neural networks, whose central premise lies in the integration of traditional Cox proportional hazards modeling with deep learning architectures. Through its multi-layered neural network structure, the algorithm autonomously learns intricate feature interactions and captures complex nonlinear relationships between covariates and survival risk. A distinctive advantage of the DeepSurv model is its capacity to recommend optimal therapeutic strategies by computationally comparing risk scores under alternative treatment regimens. While our selective SMOTE strategy improved discriminatory performance, it departs from conventional survival analysis practices and warrants further validation in external cohorts.

We note that multiple studies [50–55] have successfully integrated SMOTE and its variants (e.g. Borderline-SMOTE, SMOTE-ENN) with machine learning models—such as Random Forests, XGBoost, and Logistic Regression—in medical prediction tasks. These efforts have substantially enhanced model performance across critical domains, including cancer risk stratification (e.g. breast and prostate cancer), early detection of neurological disorders (e.g. Alzheimer's disease, delayed encephalopathy after CO poisoning), and identification of high-risk cardiovascular populations (e.g. chronic heart failure, ischemic heart disease). Empirical results demonstrate that by addressing class imbalance, SMOTE-optimized models achieve clinically meaningful improvements in key metrics (e.g. sensitivity, specificity, and AUC), thereby providing reliable support for precision medicine decision-making.

We have performed multivariable Cox regression sensitivity analysis, and the findings support surgical survival benefits across pancreatic cancer stages, particularly noting the relatively smaller benefit in Stage IV N1 subgroup (34% risk reduction). This aligns with the original conclusion that 'later-stage patients

derive diminishing surgical returns', with results now integrated into the Results and Discussion sections. SMOTE, as a classic oversampling technique for classification tasks, exhibits a theoretical conflict between its assumption of independent and identically distributed samples and the censoring mechanism prevalent in survival data. In this study, synthetic samples are generated only for the baseline covariates of deceased subjects, while the original survival time information (including censored data) is strictly preserved. This approach thus avoids the distributional distortion that would result from the direct synthesis of survival times. However, this method may not be suitable for datasets exhibiting strong time-dependent covariates or high censoring rates.

Our study findings indicate that surgical treatment can improve survival rates across all stages. SHAP-based surgical benefit assessment constitutes an individualized prediction framework integrating machine learning with clinical decision-making. This method quantifies expected survival benefit from surgery for specific patients through feature contribution decomposition: (1) constructing counterfactual scenarios to predict survival risk under surgical *versus* non-surgical management; (2) computing differences in risk scores and SHAP values; (3) outputting clinically interpretable metrics including Absolute Risk Reduction (ARR) and SHAP effect sizes. Transcending limitations of conventional staging, this approach identifies potential high-benefit candidates beyond standard criteria (e.g. oligometastatic patients) and surgery/non-surgery equivalence cohorts. It provides quantitative evidence for multidisciplinary tumor boards, enabling a paradigm shift from population-level guidelines to personalized precision recommendations.

Despite a gradual decline in survival rates with increasing stages, statistical differences persist. Recent advances have enabled pancreatic surgery to be performed safely with low morbidity and mortality rates, particularly in experienced centers where postoperative mortality rates have dropped below 5% [56,57]. Therefore, we believe there is a need to reassess the effectiveness of surgery in cases of metastatic disease, with efforts to increase resection rates potentially enhancing survival for these patients.

For stage IV patients, surgical treatment, excluding T4 and N1 stages, confers survival benefits. Tumors classified as T4 often involve critical vascular encasement, falling within a borderline resectable or unresectable range. According to predictive results, we suggest that for stage IV patients whose tumors remain resectable, primary tumor resection (PTR) may extend survival. However, the benefit of surgery is limited if stage IV patients exhibit regional lymph node metastases concurrently.

Stage T4 pancreatic cancer is characterized by tumor invasion of critical vascular structures, such as the celiac axis, superior mesenteric artery, or hepatic artery. This not only increases surgical difficulty but also reflects fundamental alterations in tumor biology. Tumor cells activate the non-canonical ALK7 pathway, inducing nuclear translocation of β-catenin, which subsequently upregulates MMP7/MMP9 expression. These Matrix Metalloproteinases (MMPs) specifically degrade key basement membrane components, including type IV collagen and laminin, thereby promoting disruption of vascular barriers and facilitating intravasation. This mechanism is markedly active in T4 tumors, resulting in potential preoperative dissemination of Circulating Tumor Cells (CTCs) even after achieving R0 resection [58]. This explains the limited survival benefit of surgery in T4 patients.

Lymph node metastasis (N1) signifies that tumor cells have acquired full metastatic competence. The underlying molecular mechanisms elucidate why surgical intervention provides limited benefit in such patients [59]. As documented in the literature, vimentin—an established epithelial-mesenchymal transition (EMT) marker—enhances tumor cell migration and invasiveness. This provides a molecular rationale for the constrained surgical efficacy in N1 patients: even after macroscopic resection of metastatic lymph nodes, tumor cells with an EMT phenotype may have already established micrometastases *via* lymphatic vessels. Concurrently, from an immunological perspective, lymph node metastasis coincides with a profoundly immunosuppressive tumor microenvironment [60].

However, this study has certain limitations. Limitations include SEER's absence of metastasis-specific descriptors (anatomic localization, disease burden, or targeted therapy for lesions) and unavoidable selection bias inherent to retrospective designs. Despite propensity score matching and multivariable adjustment, residual confounding from unmeasured factors remains possible. However, *E*-value analysis demonstrated that unmeasured confounders would require implausibly strong associations (risk ratios >2.4) to negate the observed statistical significance. Furthermore, while neural networks' opaque internal decision logic inherently limits mechanistic interpretability, we mitigated this constraint through SHAP

(SHapley Additive exPlanations) value analysis, which quantifies feature-specific contributions to individual predictions. Additionally, AI-based predictions cannot yet serve as a basis for clinical decision-making.

## 5. Conclusion

Surgical treatment reduces mortality rates for patients at different stages of pancreatic cancer, although the efficacy diminishes with later stages, it remains statistically significant. For stage IV patients, surgical treatment, excluding T4 and N1 stages, confers benefits compared to non-surgical treatments.

## Acknowledgments

Not applicable.

## Ethical approval

The data of this study comes from the SEER database. The SEER database is a tumor-related database developed by the National Cancer Institute of the United States, providing research data for researchers free of charge. All patients participating in the study received the ethical approval sought by the National Cancer Institute. The informed consent was obtained from all patients or, if patients are under 18, from a parent and/or legal guardian. The use of all data in the study follows the National Cancer Institute's data use statement. The research content follows the statement of the National Cancer Institute guidelines.

## Consent for publication

Not applicable.

## Authors contributions

CRediT: **Xin Wang**: Data curation, Writing – original draft; **Wenmao Yan**: Investigation, Methodology; **Hongyi Zhang**: Supervision, Writing – review & editing.

## Disclosure statement

No potential conflict of interest was reported by the author(s).

## Funding

None.

## Data availability statement

We obtained permission to access the database after signing and submitting the SEER Research Data Agreement form *via* email. The data that support the findings of this study are available from SEER database but restrictions apply to the availability of these data, which were used under license for the current study, and so are not publicly available. Data are however available from the authors upon reasonable request and with permission of SEER database.

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